The prompts and responses below have been edited only for formatting compatibility.

User Prompt

We work over $\mathbb{C}$. Let $\mathbb{E} \to \mathcal{M}_g$ be the rank g Hodge bundle over the moduli stack of genus g curves. Prove that $\mathbb{E}$ does not contain any non-trivial sub-vector bundles. (A sub-vector bundle is a vector bundle $V \to \mathcal{M}_g$ with an injective map $i : V \hookrightarrow \mathbb{E}$ which is injective on all fibers. A sub-vector bundle is trivial if it is either the 0-section or the whole bundle.)

Model Response (*Aletheia*)

Abstract

We prove that the rank g Hodge bundle $\mathbb{E} \to \mathcal{M}_g$ over the moduli stack of smooth complex algebraic curves of genus $g \geq 2$ admits no non-trivial proper sub-vector bundles. This is achieved by analyzing the equivariant characters of automorphisms on putative subbundles over specifically constructed loci of curves with a $(\mathbb{Z}/2\mathbb{Z})^2$-action. By extracting a rigid functional equation for the character traces, we categorically force the dimension of the isotypic invariant subspaces to be non-integral, yielding an algebraic contradiction.

1 Introduction

Let $\mathcal{M}_g$ be the moduli stack of smooth projective curves of genus $g \geq 2$ over $\mathbb{C}$. The Hodge bundle $\mathbb{E} \to \mathcal{M}_g$ is the rank g vector bundle whose fiber over a moduli point $[C]$ is the space of holomorphic abelian differentials $H^0(C, \omega_C)$.
Suppose for the sake of contradiction that there exists a globally defined proper non-trivial sub-vector bundle $V \subset \mathbb{E}$ of rank r. By definition, this means $V \to \mathcal{M}_g$ admits a bundle map $V \hookrightarrow \mathbb{E}$ that is injective on all fibers, and its rank satisfies $0 < r < g$. For any curve $C \in \mathcal{M}_g$, the fiber V_C is an r-dimensional subspace of $H^0(C, \omega_C)$ which is naturally invariant under the pullback action of the full automorphism group $\mathrm{Aut}(C)$.

2 Equivariant Traces of Involutions

Lemma 1. *Let τ be an orientation-preserving involution on a smooth curve C of genus g with $2m$ fixed points. The trace of the pullback τ^* on V_C is a constant integer $f(m)$ depending only on m and the bundle V.*

Proof. By the topological classification of surfaces, two orientation-preserving involutions on a closed orientable surface Σ_g with the same number of fixed points correspond to double covers over a quotient surface of a uniquely determined genus. Thus, such involutions are conjugate in the mapping class group Mod_g. The putative bundle V functorially corresponds to a Mod_g-equivariant holomorphic vector bundle over the Teichmüller space $\mathcal{T}_g$.

By the Nielsen Realization Theorem (S. P. Kerckhoff, *The Nielsen Realization Problem*, Ann. of Math., 1983), the fixed-point locus $\mathcal{T}_g^\tau$ of τ is non-empty and naturally identifies with the Teichmüller space of the quotient orbifold $\Sigma_g/\langle\tau\rangle$. Since the Teichmüller space of any orbifold is homeomorphic to a Euclidean space, $\mathcal{T}_g^\tau$ is a connected manifold. The restriction of the equivariant bundle V to $\mathcal{T}_g^\tau$ globally splits into $+1$ and -1 eigenbundles for τ^*. Because $\mathcal{T}_g^\tau$ is connected, the ranks of these eigenbundles are strictly constant. The trace of τ^* on V_X for any $X \in \mathcal{T}_g^\tau$ is exactly the difference of these constant ranks, forming an integer $f(m)$ dictated entirely by the conjugacy class of τ, and hence uniquely determined by m. $\square$

By the Riemann-Hurwitz formula applied to the quotient map $C \to C/\langle\tau\rangle$, we have $2g-2 = 2(2g'-2)+2m$, meaning $m = g+1-2g'$. Since the quotient genus g' is a non-negative integer, the geometrically valid parameters m lie strictly in the finite set:

$$S = \{m \in \mathbb{Z} \mid 0 \le m \le g+1,\ m \equiv g+1 \pmod 2\}.$$

3 A Functional Equation via $(\mathbb{Z}/2\mathbb{Z})^2$-Covers

Lemma 2. *For any three integers $m_1, m_2, m_3 \in S$ such that $m_1+m_2+m_3 = g+3$ and at most one $m_i = 0$, we have the uniform trace relation $f(m_1)+f(m_2)+f(m_3) = -r$.*

Proof. Given a valid triple (m_1, m_2, m_3), we choose generic, mutually coprime, square-free polynomials $A(x), B(x), C(x)$ over $\mathbb{C}$ of degrees m_1, m_2, m_3, respectively. Let $K = \mathbb{C}(x)$. We define a smooth projective curve C as the smooth model of the function field $L = K(y,z)$ where $y^2 = A(x)C(x)$ and $z^2 = B(x)C(x)$. Because at most one $m_i = 0$, the polynomials cannot all be constants; thus the extension L/K forms a fully generated degree 4 Galois extension with Galois group $G \cong (\mathbb{Z}/2\mathbb{Z})^2$, guaranteeing C is a connected curve. Since $m_i \equiv g+1 \pmod 2$, the degrees of A, B, and C strictly share the same parity. Consequently, the polynomials $A(x)C(x)$ and $B(x)C(x)$ possess even degrees, meaning L/K is unramified over the place at $x = \infty$. By local Galois analysis, the ramification of L/K is localized entirely over the roots of A, B, and C:

- Over a root of $A(x)$, the subextension $K(z)/K$ is unramified (as B and C do not vanish there), naturally splitting into 2 places in $K(z)$. Adjoining y introduces ramification of index 2 at both places. Thus, there are 2 places in L, both with ramification index 2 over K.
- Symmetrically, there are exactly 2 places over each root of $B(x)$, each with ramification index 2 over K.
- Over a root c of $C(x)$, the subextension $K(y/z)/K$ evaluates locally to $(y/z)^2 = A(c)/B(c) \neq 0$, which cleanly splits into 2 places over $\mathbb{C}$. Adjoining y (and therefore z) introduces ramification of index 2 at both places.

By the Riemann-Hurwitz formula applied to L/K, the genus g_C satisfies:

$$2g_C - 2 = 4(-2) + 2m_1 + 2m_2 + 2m_3 = -8 + 2(g+3) = 2g-2,$$

which formally confirms $g_C = g$.
The curve C admits an automorphism group $G \cong (\mathbb{Z}/2\mathbb{Z})^2$ generated by the involutions $\tau_1(y,z) = (-y,z)$ and $\tau_2(y,z) = (y,-z)$, with $\tau_3 = \tau_1\tau_2 = (-y,-z)$. The fixed points of τ_1 correspond precisely to the places of L fixed by the involution τ_1, meaning τ_1 lies in their inertia group. Because $L = K(z)(y)$ and $y^2 = A(x)C(x)$, ramification in $L/K(z)$ occurs exactly where the local valuation of $A(x)C(x)$ in $K(z)$ is odd.
Over a root of $A(x)$, $A(x)C(x)$ has a simple zero in K, and since $K(z)/K$ is unramified here, the valuation in $K(z)$ remains 1 (odd). Thus, these places strictly ramify in $L/K(z)$, meaning τ_1 generates the inertia group and fixes the places. As there are 2 places in $K(z)$ over each of the m_1 roots of $A(x)$, this contributes exactly $2m_1$ fixed points.
Conversely, at a place of $K(z)$ lying over a root c of $C(x)$, the local affine parameter for the base field $K = \mathbb{C}(x)$ is $x - c$. We have $z^2 = B(x)C(x)$. Since $B(c) \neq 0$ and $C(x)$ has a simple root at c, z^2 vanishes to order 1 with respect to $x - c$, making z a local uniformizer for $K(z)$ at this unique place. Consequently, $x - c$ vanishes to order 2 with respect to z, and thus $A(x)C(x)$ also inherently vanishes to order 2. Since its valuation is even, $(y/z)^2$ evaluates locally to a non-zero complex number $A(c)/B(c)$. Extracting the square root cleanly splits into two unramified places in L. Because they are unramified, the non-trivial automorphism τ_1 of $L/K(z)$ simply swaps these two places, yielding zero fixed points.
Finally, over a root of $B(x)$, the field $K(y)$ is unramified, meaning there are 2 places in $K(y)$ where y takes distinct non-zero values. The involution τ_1 maps $y \mapsto -y$, explicitly swapping these places, thus fixing none.
Therefore, τ_1 possesses exactly $2m_1$ fixed points. By identical symmetry, τ_2 and τ_3 have exactly $2m_2$ and $2m_3$ fixed points, respectively.
The space V_C decomposes into four G-isotypic components: $V^{++}, V^{+-}, V^{-+}, V^{--}$. Projecting onto the invariant subspace via character theory yields:

$$\dim V^{++} = \frac{1}{4}\big(\operatorname{tr}(1) + \operatorname{tr}(\tau_1) + \operatorname{tr}(\tau_2) + \operatorname{tr}(\tau_3)\big) = \frac{1}{4}\big(r + f(m_1) + f(m_2) + f(m_3)\big).$$

Crucially, V^{++} is a subspace of the G-invariant differentials $H^0(C, \omega_C)^G$, which canonically correspond to holomorphic differentials on the quotient curve C/G. Since the invariant field is exactly $K = \mathbb{C}(x)$, the quotient curve is natively $\mathbb{P}^1$. Because $H^0(\mathbb{P}^1, \omega_{\mathbb{P}^1}) = 0$, we unconditionally deduce $\dim V^{++} = 0$. This forces the functional equation $f(m_1) + f(m_2) + f(m_3) = -r$. □

4 Solving the Character Recurrence

Lemma 3. *For all $m \in S$, the unique solution to the functional equation is $f(m) = \frac{r(1-m)}{g}$.*

Proof. Let $y(m) = f(m) - \frac{r(1-m)}{g}$. Since the affine expression $\frac{r(1-m)}{g}$ algebraically satisfies the sum relation for any triple summing to $g+3$, we immediately deduce the homogeneous equation $y(m_1) + y(m_2) + y(m_3) = 0$ for all valid triples.

When $m = g+1$, the involution possesses $2g+2$ fixed points, uniquely identifying it as the hyperelliptic involution. The hyperelliptic involution universally acts as $-I$ on the canonical bundle, thus acting as $-I$ on $H^0(C, \omega_C)$ and its subspace V_C. Hence, $f(g+1) = -r$, enforcing the strong boundary condition $y(g+1) = 0$.

Case g is even: The set S consists of odd integers $m_i = 2k_i + 1$ with $0 \leq k_i \leq g/2 = n$. Since $m_i \geq 1$, no valid index is zero. Let $y_k = y(2k+1)$. The relation for the index triple $(k_1, k_2, k_3) = (0, n, 0)$ corresponds to the valid m-values $(1, g+1, 1)$, which cleanly sum to $g+3$. This yields $2y_0 + y_n = 0 \implies y_0 = 0$. For any $0 \leq j \leq n-1$, the index triples $(1, j, n-1-j)$ and $(0, j+1, n-1-j)$ correspond to valid m-triples summing to $g+3$. Subtracting their respective sum equations isolates $y_{j+1} - y_j = y_1$, forming an arithmetic progression $y_k = ky_1$. Because $y_n = 0$ and $n \geq 1$, we firmly deduce $y_k = 0$ for all k.

Case g is odd: The set S consists of even integers $m_i = 2k_i$ with $0 \leq k_i \leq (g+1)/2 = n-1$, where $n = (g+3)/2$. Let $y_k = y(2k)$. The boundary condition guarantees $y_{n-1} = 0$. For $1 \leq k \leq n-2$, the index triples $(1, k, n-1-k)$ and $(0, k+1, n-1-k)$ correspond to m-triples summing to $g+3$, each structurally containing at most one zero index. Subtracting their respective equations gives $y_{k+1} - y_k = y_1 - y_0$, establishing an arithmetic progression. The valid index triple $(0, 1, n-1)$ corresponds to m-values $(0, 2, g+1)$, safely possessing exactly one zero, and yields $y_0 + y_1 + y_{n-1} = 0 \implies y_1 = -y_0$. The progression thus evaluates to $y_k = y_0(1-2k)$. At $k = n-1$, $y_{n-1} = y_0(1 - 2(n-1)) = y_0(3-2n) = 0$. Since $g \geq 3$, we have $3 - 2n = -g \neq 0$, forcing $y_0 = 0$, meaning $y_k = 0$ for all k. □

5 The Integrality Contradiction

Theorem 1. *The Hodge bundle $\mathbb{E} \to \mathcal{M}_g$ contains no non-trivial proper sub-vector bundles.*

Proof. Isolating the trace projection for the V^{+-} eigenspace yields:

$$\dim V^{+-} = \frac{1}{4}\big(\operatorname{tr}(1) + \operatorname{tr}(\tau_1) - \operatorname{tr}(\tau_2) - \operatorname{tr}(\tau_3)\big) = \frac{1}{4}\big(r + f(m_1) - f(m_2) - f(m_3)\big).$$

Substituting the algebraic identity $-f(m_2) - f(m_3) = r + f(m_1)$ via Lemma 2 cleanly simplifies this structure to:

$$\dim V^{+-} = \frac{1}{2}\big(r + f(m_1)\big).$$

Consequently, $2\dim V^{+-} = r + f(m_1) = r + \frac{r(1-m_1)}{g}$. Because $\dim V^{+-}$ represents the formal dimension of a complex vector space, $2\dim V^{+-}$ must intrinsically be an even integer. Thus, the evaluated expression $r + \frac{r(1-m_1)}{g}$ is restricted to be an **even integer** for all values $m_1 \in S$ capable of completing a valid triple.

- **If g is even:** Both 1 and 3 inherently belong to S. They seamlessly extend to the valid triples $(1, 1, g+1)$ and $(3, 1, g-1)$ respectively, summing to $g+3$ with no zero entries. Evaluating at $m_1 = 1$ strictly produces r, logically demanding

that r is an even integer. Evaluating at $m_1 = 3$ yields $r - \frac{2r}{g}$, which must simultaneously act as an even integer.

- **If g is odd:** Both 0 and 2 inherently belong to S. They smoothly extend to the valid triples $(0, 2, g+1)$ and $(2, 2, g-1)$ respectively, summing to $g+3$ with at most one zero entry (since $g \geq 3 \implies g - 1 \geq 2$). Evaluating at $m_1 = 0$ yields $r + \frac{r}{g}$, which is forced to be an even integer. Evaluating at $m_1 = 2$ yields $r - \frac{r}{g}$, which must similarly act as an even integer.

In either topological scenario, calculating the arithmetic difference between the evaluations at the two adjacent valid states strictly isolates $\frac{2r}{g}$. Because the difference between any two even integers categorically generates another even integer, we deduce:

$$\frac{2r}{g} \in 2\mathbb{Z} \implies \frac{r}{g} \in \mathbb{Z} \implies r \equiv 0 \pmod{g}.$$

Since V was hypothesized to be a proper subbundle, its rank is structurally constrained to the open interval $0 < r < g$. However, the parity condition categorically demands that r behaves as a positive integer multiple of g. As absolutely no multiple of g exists strictly within the open interval $(0, g)$, we achieve an absolute mathematical contradiction.

Hence, no non-trivial proper sub-vector bundle $V \subset \mathbb{E}$ can topologically or algebraically exist over $\mathcal{M}_g$. $\square$